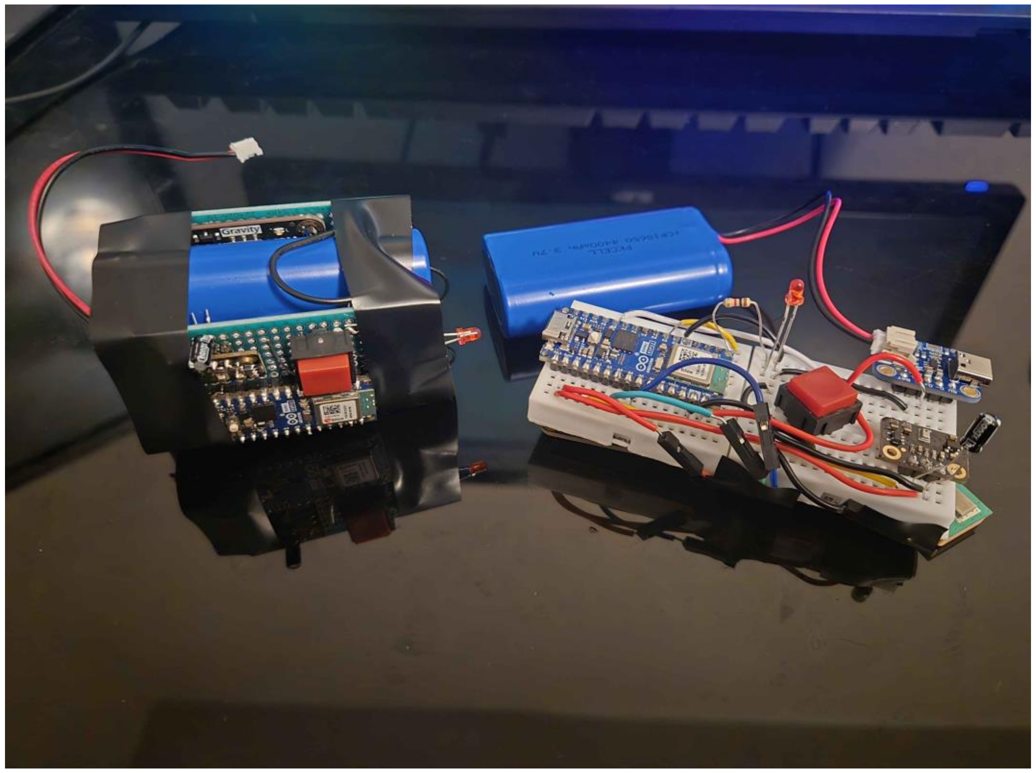
CCEW WAD Final Demo

ECE Capstone Final Presentation

By: Jonas W, Cody C, Alex H, Ryan K

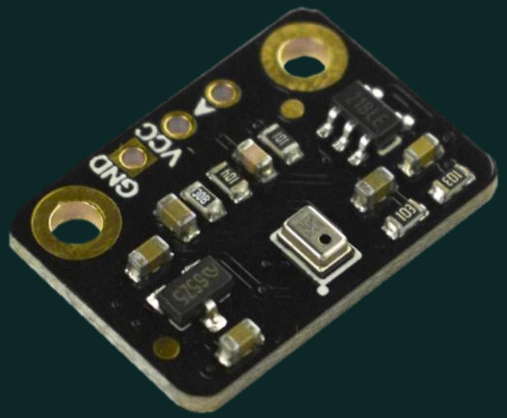
Overview



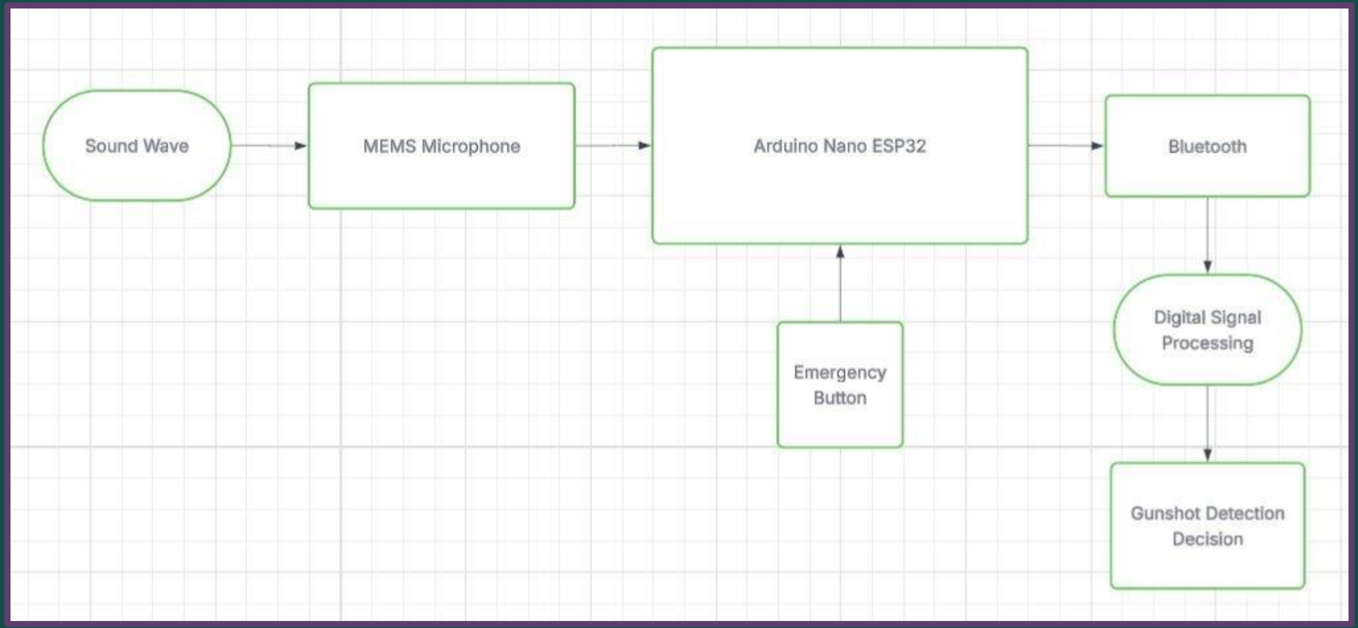
- The Mission of the WAD
  - A wearable device that detects emergencies and sends real-time location, elevation, and threat data to authorities.
- The Intended Audience of the WAD
  - Secondary responders (e.g., security, maintenance, social workers) who may face risk without formal emergency training.

Emergency Awareness, Instantly Delivered

Specific Features



- Purpose:
  - Detect potential gunshots in real time
  - Alert emergency response with button or gunshot
- How does it work?



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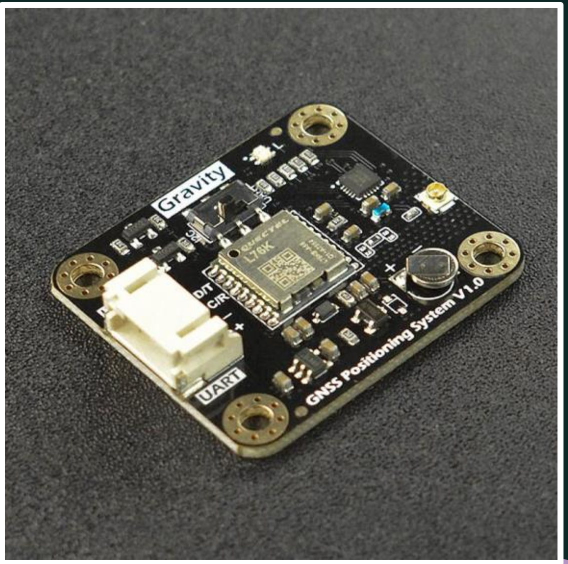
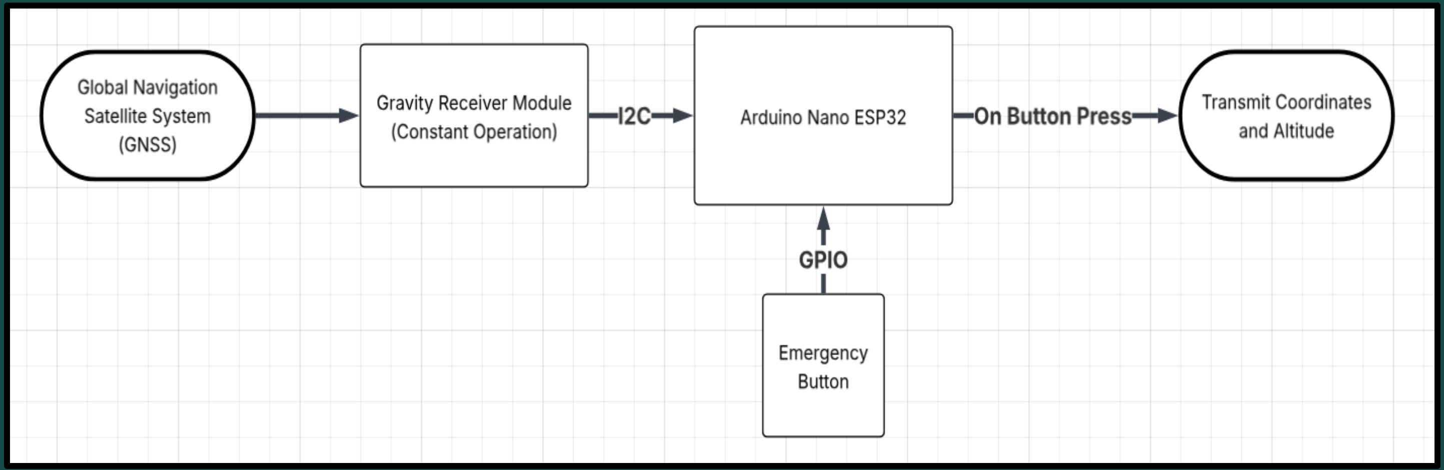
Specific Features Con't



- Purpose:
  - Time-stamps coordinates and altitude of device constantly
- How does it work?



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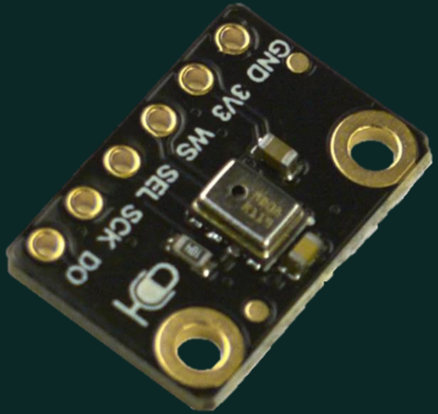
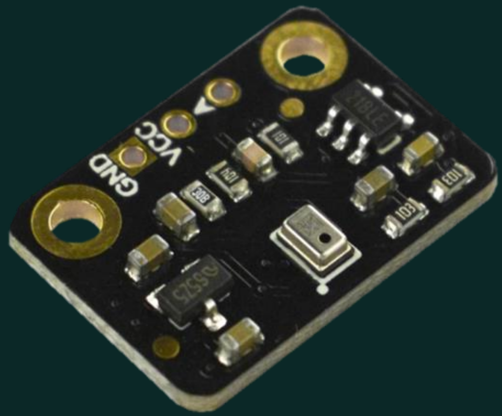
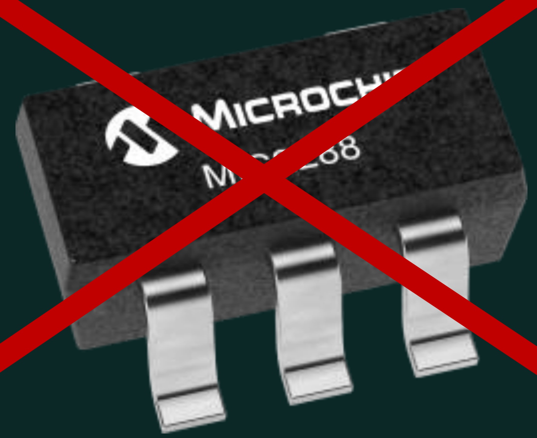
Changes Made in Project



Fermion: I2S MEMS Microphone



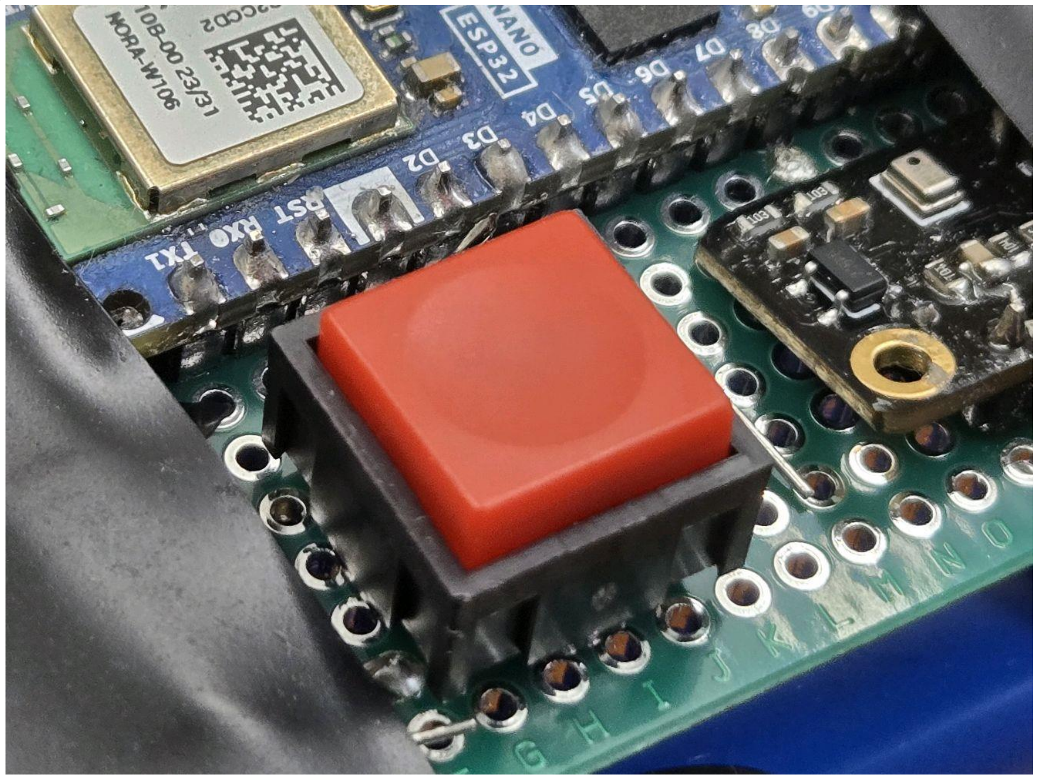
Fermion: MEMS Microphone Module



Testing and Demonstration: Button and Wireless Data Transmission

Wireless data transmission occurs in two main scenarios:

1. User presses the button
2. Gunshot is detected



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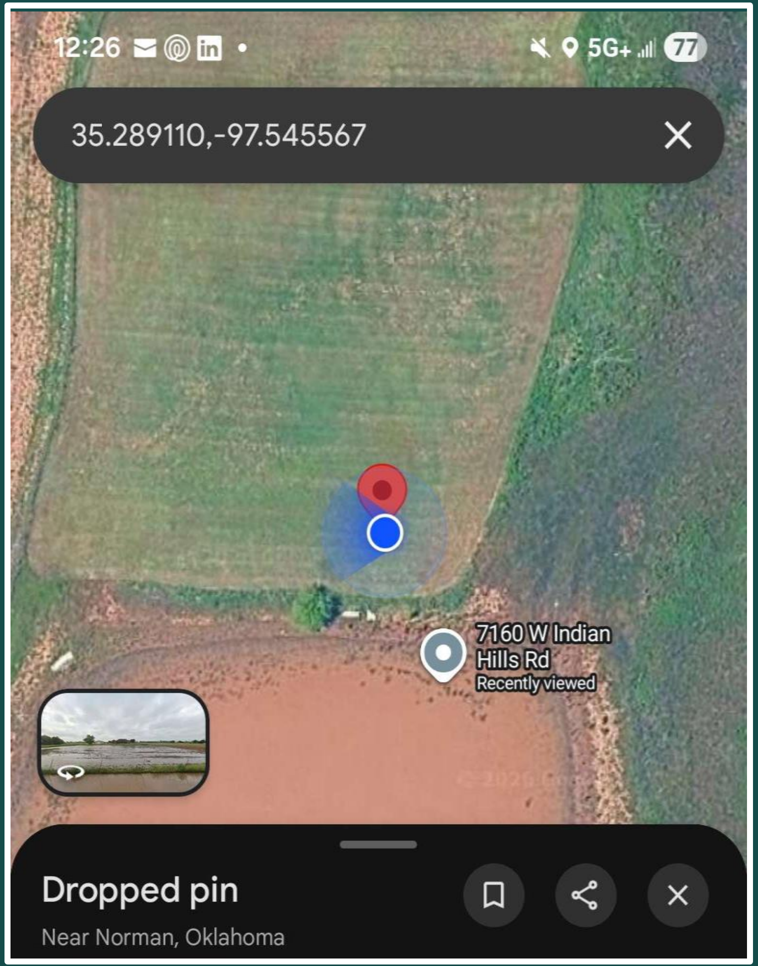
Testing and Demonstration: GPS Module

Location: Farm



Percent Differences:
(calculations in Appendix C)

Latitude: $\sim 2.8 * 10^{-4}$ %

Longitude: $\sim 2.7 * 10^{-4}$ %

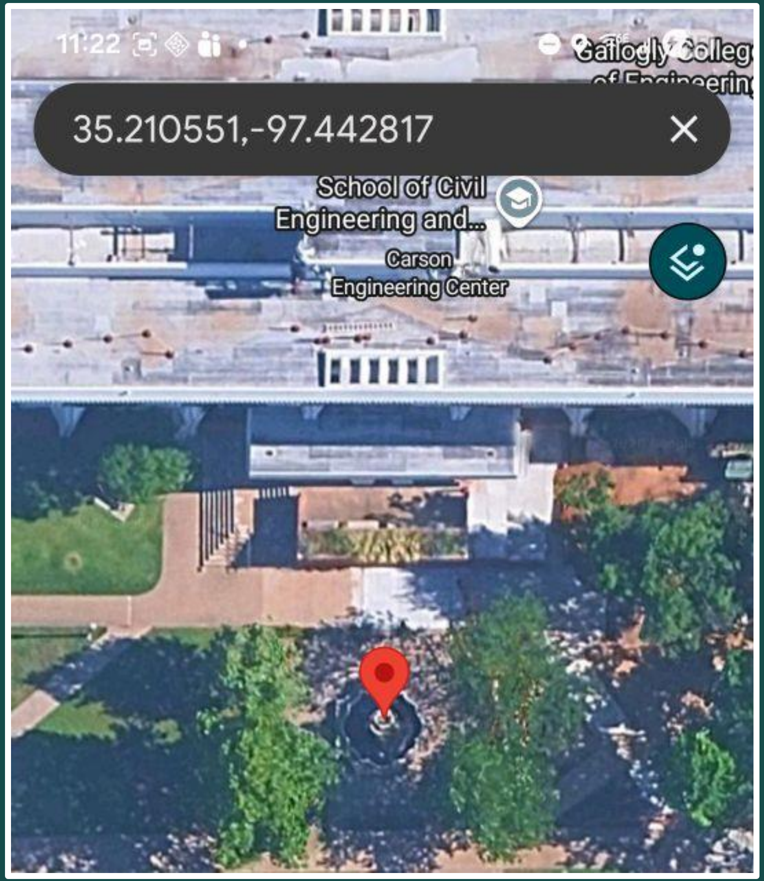
Altitude: ~ 0.378 %

Mobile App Data	Value
Timestamp (CST)	4/24/2026 12:26:38 PM
Latitude (Degrees)	35.28901°
Longitude (Degrees)	-97.54530°
Altitude (Meters)	345.69 m

AH/
RK

Testing and Demonstration: GPS Module

Location: OU Engineering Quad Fountain



Percent Differences:
(calculations in Appendix C)

Latitude: $\sim 1.1 * 10^{-4}$ %

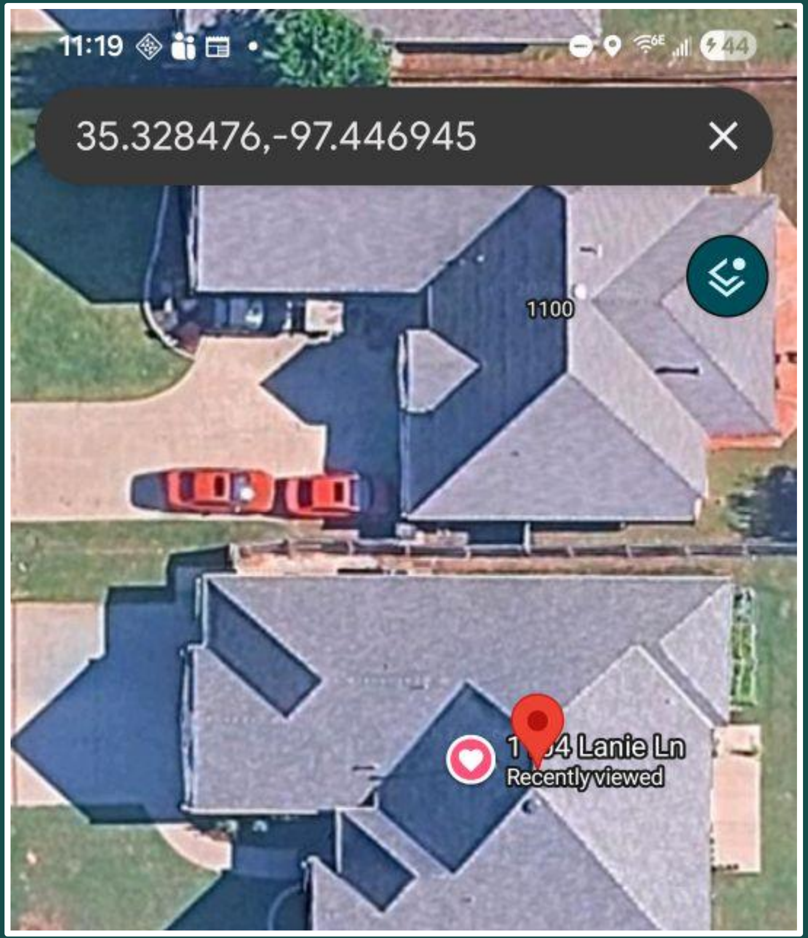
Longitude: $\sim 2.8 * 10^{-5}$ %

Altitude: ~ 3.040 %

Mobile App Data	Value
Timestamp (CST)	4/29/2026 10:17:14 PM
Latitude (Degrees)	35.21059°
Longitude (Degrees)	-97.44279°
Altitude (Meters)	350.19 m

Testing and Demonstration: GPS Module

Location: Alex's Home



Percent Differences:
(calculations in Appendix C)

Latitude: ~$5.3 * 10^{-4}$ %

Longitude: ~$6.2 * 10^{-6}$ %

Altitude: ~6.703 %

Mobile App Data	Value
Timestamp (CST)	4/3/26 20:20:23 PM
Latitude (Degrees)	35.328664°
Longitude (Degrees)	-97.446954°
Altitude (Meters)	397.80 m

Testing and Demonstration: Gunshot Detection

Brief Overview of Detection Process

Note: For more details, full software block diagram available in Appendix A

- Test #1: Detect audio spikes (in dB) above a threshold. → Test #2: Find strong gunshot frequencies (in Hertz) below a certain threshold. → If both tests pass, then a gunshot has been detected.

Audio Spike Detection Test

Frequency Detection Test

Testing and Demonstration: Gunshot Detection

Firearm: Glock 17

Gunshot Audio Spike Threshold (dB): 80
Gunshot Frequency Threshold (Hz): 1500

Total Number of Gunshots	Audio Spikes Detected	Gunshot Frequency Detected	Gunshots Detected
0	0	0	0

Mobile App Data	Value
Timestamp (CST)	4/24/2026 12:50:34 PM
Latitude (Degrees)	35.28904°
Longitude (Degrees)	-97.54551°
Altitude (Meters)	334.80 m
Max dB Reading (dB SPL)	79 dB SPL
Gunshot Detected?	False
Call Emergency Responders?	True

WAD Service
UUID: 3098a182-02b0-4a35-889a-93b71e1748b9
PRIMARY SERVICE

WAD Characteristic
UUID: b536bcce-2c24-4f7c-bf9f-1de558316263
Properties: NOTIFY, READ
Value: {"timestamp_cst":"2026-04-24 12:50:34","lat_degrees:":35.28904,"long_degrees:":-97.54551,"alt_meters:":334.80,"max_dB_reading:":79,"gunshot_detected":false,"call_responders":false}
Descriptors:
Client Characteristic Configuration
UUID: 0x2902
Value: Notifications enabled

Testing and Demonstration: Gunshot Detection

Firearm: AR-15

Gunshot Audio Spike Threshold (dB): 80
Gunshot Frequency Threshold (Hz): 1500

Total Number of Gunshots	Audio Spikes Detected	Gunshot Frequency Detected	Gunshots Detected
0	0	0	0

Mobile App Data	Value
Timestamp (CST)	4/24/2026 12:56:37 PM
Latitude (Degrees)	35.28907°
Longitude (Degrees)	-97.54552°
Altitude (Meters)	354.30 m
Max dB Reading (dB SPL)	70 dB SPL
Gunshot Detected?	False
Call Emergency Responders?	True

WAD Service
UUID: 3098a182-02b0-4a35-889a-93b71e1748b9
PRIMARY SERVICE

WAD Characteristic
UUID: b536bcce-2c24-4f7c-bf9f-1de558316263
Properties: NOTIFY, READ
Value: {"timestamp_cst":"2026-04-24 12:56:05","lat_degrees:":35.28915,"long_degrees:":-97.54553,"alt_meters:":356.39,"max_dB_reading:":83,"gunshot_detected":true,"call_responders":true}
Descriptors:
Client Characteristic Configuration
UUID: 0x2902
Value: Notifications enabled

Testing and Demonstration: Gunshot Detection

Conclusions

Glock 17: $Accuracy \approx 60\%$

AR-15: $Accuracy \approx 83.3\%$

(for calculations, see Appendix D)

- Audio spike detection worked 100% of the time
- If frequency detection was removed from software, gunshots would be detected 100% of the time

Final Budget

Item	Item Description	Quantity	Cost $	Use	Part Number
1	Arduino Nano ESP32	2	$38.60	Main Micro controller for project	Mouser - ABX00083
2	Lithium Ion Battery Pack - 3.7V 4400mAh	2	$39.90	Main Power supply for project	Adafruit - 354
3	Micro-Lipo Charger for LiPoly Batt with USB Type C Jack	2	$11.90	Rechargable module for Battery Pack	Adafruit - 4410
4	Gravity: GNSS GPS BeiDou Receiver Module - I2C&UART	2	$35.80	GPS Module for Project	Digikey - 1738-TEL0157-ND
5	FERMION MEMS MICROPHONE MODULE(Analog)	2	$5.80	Microphone Module for GS Detection	Digikey - SEN0487
6	320E11RED	2	$4.18	Button for panic device	Digikey - 101391
7	MIC2288YD5-TR	2	$1.46	Booster DC DC Converter	Mouser - MIC2288YD5
8	Breakout Board	2	$2.50	Breakout board for Booster converter	Digikey - 1568-00717-ND
9	SCHOTTKY RECTIFIER	3	$0.87	Schottky diode for GPS Module	Digikey -497-11370-1-ND - Cut Tape (CT)
10	Testing LED (Red)	1	$0.55	LED used to tell user if device is working	Digikey - SLR-56VC3F
11	Decupling Capacitor (0.1 Microfarad)	1	$0.89	Used to remove noise from battery pack power	Digikey - C322C104K3G5TA
12	Current Limiting Resistor(1 KOhm)	1	$0.10	Prevents blow out of LED	Digikey - CFF14JT1K00
13	Wires(Jumper/Cut) (Spool)	1	$15.89	Came with boards but used to connect device	Half came with the Micro breadboards/ other half from link
14	Micro Breadboards	2	$1.98	Used to seat device	6 Pcs Mini Small Breadboard kit White 170 Tie Points
15	Proto Boards	2	$0.56	Used to more perminently seat device	ELEGOO 32 Pcs Double Sided PCB Board Prototype Kit
16	Electrical Tape(Roll)	1	$0.97	Used to hold device together before case	Hyper Tough Electrical Tape Single Roll 60 Feet Long Black
		Total:	$161.95		

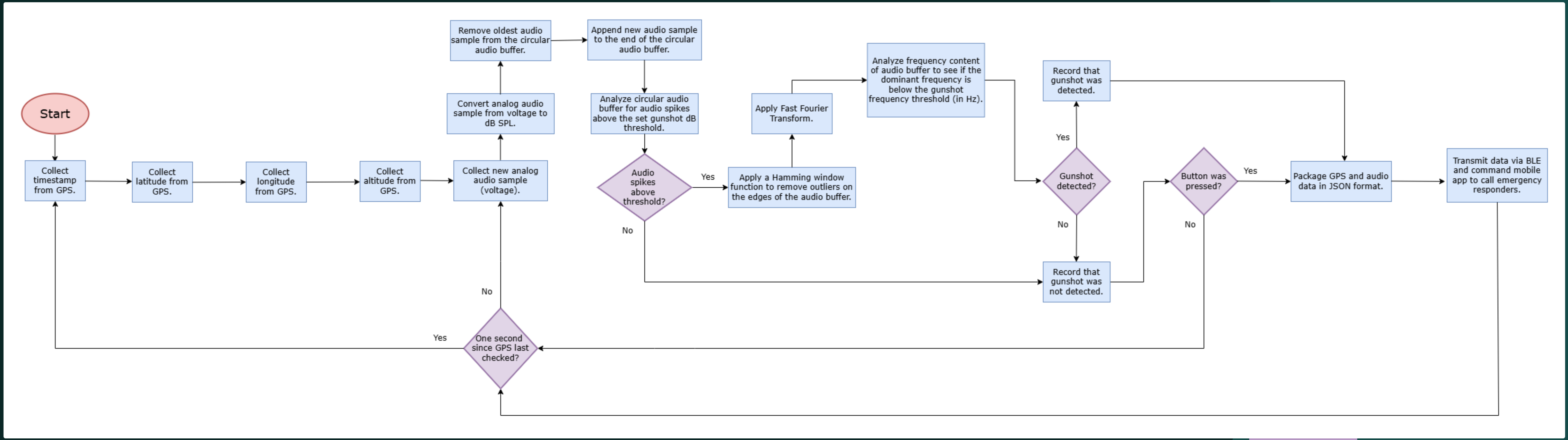
Future Plans

- **Hardware Future Plans:**
  - Higher Decibel Microphone
  - PCB Implementation
  - Smaller form factor
  - Off-On Switch
  - Extending GPS signal range
  - Vibration feature
- **Software Future Plans:**
  - Smarter gunshot pattern detection
  - Refining software to avoid slowing down audio sampling
  - User interface to change internal settings of the WAD

Questions?

Appendix A

Software Block Diagram



Appendix B

Battery Calculations

- **Battery Life:**

$$h = \frac{C}{I_{avg}}$$

$$h = \frac{4400\ mAh}{167\ mA}$$

$$h \approx 26.35\ Hours$$

h – *Battery Life in Hours*

C – *Battery Capacity (mAh)*

I_{avg} – *average current of entire system*

- **Charge Time:**

$$t = \frac{C}{A}$$

$$t = \frac{4400mAh}{100mA}$$

$$t = 44\ Hours$$

t – *Charge Time*

C – *Battery Capacity (mAh)*

A – *Charge Current*

AH

Appendix C

GPS Tracking Percent Difference Calculations

Farm:

Latitude:

$$Percent\ Difference = \left|\frac{W_{lat} - M_{lat}}{\frac{W_{lat} + M_{lat}}{2}}\right| * 100\% \approx 2.8 * 10^{-4}\ \%$$

Longitude:

$$Percent\ Difference = \left|\frac{W_{long} - M_{long}}{\frac{W_{long} + M_{long}}{2}}\right| * 100\% \approx 2.7 * 10^{-4}\ \%$$

Altitude:

$$Percent\ Difference = \left|\frac{W_{alt} - M_{alt}}{\frac{W_{alt} + M_{alt}}{2}}\right| * 100\% \approx 0.378\ \%$$

Let W_{lat} represent the latitude obtained from the WAD. Let W_{long} represent the longitude obtained from the WAD. Let W_{alt} represent the latitude obtained from the WAD.

$W_{lat} \approx 35.28901°$
$W_{long} \approx -97.54530°$
$W_{alt} \approx 345.69\ m$

Let M_{lat} represent the latitude obtained from a mobile device. Let M_{long} represent the latitude obtained from a mobile device. Let M_{alt} represent the latitude obtained from a mobile device.

$M_{lat} \approx 35.289110°$
$M_{long} \approx -97.545567°$
$M_{alt} \approx 347\ m$

AH

Appendix C

GPS Tracking Percent Difference Calculations

OU Engineering Quad Fountain:

Latitude:

$$Percent\ Difference = \left|\frac{W_{lat} - M_{lat}}{\frac{W_{lat} + M_{lat}}{2}}\right| * 100\% \approx 1.1 * 10^{-4}\ \%$$

Longitude:

$$Percent\ Difference = \left|\frac{W_{long} - M_{long}}{\frac{W_{long} + M_{long}}{2}}\right| * 100\% \approx 2.8 * 10^{-5}\ \%$$

Altitude:

$$Percent\ Difference = \left|\frac{W_{alt} - M_{alt}}{\frac{W_{alt} + M_{alt}}{2}}\right| * 100\% \approx 3.040\ \%$$

Let W_{lat} represent the latitude obtained from the WAD. Let W_{long} represent the longitude obtained from the WAD. Let W_{alt} represent the latitude obtained from the WAD.

$W_{lat} \approx 35.21059°$
$W_{long} \approx -97.44279°$
$W_{alt} \approx 350.19\ m$

Let M_{lat} represent the latitude obtained from a mobile device. Let M_{long} represent the latitude obtained from a mobile device. Let M_{alt} represent the latitude obtained from a mobile device.

$M_{lat} \approx 35.210551°$
$M_{long} \approx -97.442817°$
$M_{alt} \approx 361\ m$

Appendix C

GPS Tracking Percent Difference Calculations

Alex's Home:

Latitude:

$$Percent\ Difference = \left|\frac{W_{lat} - M_{lat}}{\frac{W_{lat} + M_{lat}}{2}}\right| * 100\% \approx 5.3 * 10^{-4}\ \%$$

Longitude:

$$Percent\ Difference = \left|\frac{W_{long} - M_{long}}{\frac{W_{long} + M_{long}}{2}}\right| * 100\% \approx 6.2 * 10^{-6}\ \%$$

Altitude:

$$Percent\ Difference = \left|\frac{W_{alt} - M_{alt}}{\frac{W_{alt} + M_{alt}}{2}}\right| * 100\% \approx 6.703\ \%$$

Let W_{lat} represent the latitude obtained from the WAD. Let W_{long} represent the longitude obtained from the WAD. Let W_{alt} represent the latitude obtained from the WAD.

$W_{lat} \approx 35.328664°$
$W_{long} \approx -97.446954°$
$W_{alt} \approx 397.80\ m$

Let M_{lat} represent the latitude obtained from a mobile device. Let M_{long} represent the latitude obtained from a mobile device. Let M_{alt} represent the latitude obtained from a mobile device.

$M_{lat} \approx 35.328476°$
$M_{long} \approx -97.446945°$
$M_{alt} \approx 372\ m$

Appendix D

Gunshot Accuracy Calculations

Glock 17:

$$Accuracy = \frac{\#\ Gunshots\ Detected}{Total\ \#\ of\ Gunshots} = \frac{3}{5} = \mathbf{60\%}$$

AR-15:

$$Accuracy = \frac{\#\ Gunshots\ Detected}{Total\ \#\ of\ Gunshots} = \frac{5}{6} \approx \mathbf{83.333\%}$$